

CS COMPUTER NETWORKS ASSIGNMENT

Topic: 802.11 MAC Header Decoding & Analysis

Name: [Your Name] | **Roll No:** [Your Roll Number]

1. 802.11 MAC HEADER DECODER

1.1 Frame Control Field (2 Bytes)

First Byte:

Bits	Meaning	Values
0–1	Protocol Version	Normally 00
2–3	Type	00 = Management, 01 = Control, 10 = Data
4–7	Subtype	Depends on the frame type

Second Byte:

Bit	Flag	Meaning
0	ToDS	Frame is going toward the Distribution System
1	FromDS	Frame is coming from the Distribution System
2	More Frag	More fragments of the frame follow
3	Retry	Frame is a retransmission
4	Power Mgmt	Station's power-management state
5	More Data	More buffered data is available
6	Protected	Frame body is protected/encrypted
7	Order	Strict ordering requested

1.2 Control Frame Subtypes

Subtype bits	Frame	Address Details
1011	RTS	Contains two addresses
1100	CTS	Contains one address
1101	ACK	Contains one address

2. DURATION/ID (2 BYTES)

For the frames in this assignment, it represents the duration in microseconds for which the medium is reserved. The field is stored in little-endian order.

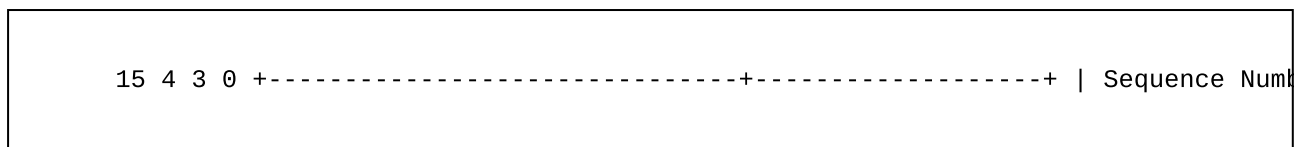
For example: 2C 00 is 0x002C = 44 μ s.

3. ADDRESS FIELDS

802.11 frames can contain different numbers of addresses depending on the frame type and ToDS/FromDS bits.

~~Address roles are defined as: RA = Receiver Address, TA = Transmitter Address, DA = Destination Address, SA =~~

4. SEQUENCE CONTROL (2 BYTES)



The Sequence Control field is also stored in little-endian order.

Example: F1 contains 30 1D which gives $0 \times 1D30 = 0001\ 1101\ 0011\ 0000$.

- Sequence Number = $0001\ 1101\ 0011 = 467$
- Fragment Number = $0000 = 0$

5. MANUAL DECODING TABLE

Field	F1	F3
Frame Control binary	00001000 00000001	10110100 00000000
Type / Subtype	Data / Data (0000)	Control / RTS (1011)
ToDS, FromDS	1, 0	0, 0
Other flags set	ToDS only	None
Duration (µs)	44	350
Address 1 (role)	A4:83:E7:11:22:33 (RA)	A4:83:E7:11:22:33 (RA)
Address 2 (role)	00:50:56:AA:BB:CC (TA/SA)	00:50:56:AA:BB:CC (TA)
Address 3 (role)	00:1A:2B:3C:4D:5E (DA)	Not present
Sequence no. / fragment no.	467 / 0	Not present

6. INTERPRETATION QUESTIONS

Q1. F3, F4 and F5 are one exchange. Name the sender and receiver of each, and explain why the Duration drops from F3 to F4 to F5.

F3 (RTS): Sender (TA) = 00:50:56:AA:BB:CC, Receiver (RA) = A4:83:E7:11:22:33, Duration = 350 µs.

F4 (CTS): Sender = A4:83:E7:11:22:33, Receiver = 00:50:56:AA:BB:CC, Duration = 324 µs.

F5 (ACK): Sender = A4:83:E7:11:22:33, Receiver = 00:50:56:AA:BB:CC, Duration = 0 µs.

Explanation: The duration field reserves the medium for the remaining time required to complete the exchange. As time passes and parts of the frame transmission occur, the remaining required time decreases, which causes the Duration value to drop from F3 to F4 to F5.

Q2. A station that hears only F4 sets its NAV. For how many microseconds, and which slide mechanism does this implement?

F4 has Duration = 324 µs. A station hearing F4 sets its NAV for **324 µs**. This implements **virtual carrier sensing using the Network Allocation Vector (NAV)**, allowing stations to defer transmissions without physical signal detection.

Q3. F2 has the Retry flag set. How does the receiver use the Retry bit together with the sequence number to discard duplicates?